Student Performance Analytics System

Project Documentation | Python with AI

Prepared from the supplied assignment requirements. Complete the student details and add actual output screenshots and GitHub URL before submitting.

1. Project Title

Student Performance Analytics System

Student Name: _____

Roll Number: _____

Department/Class: _____

Course: Python with AI

2. Objective

The objective of this project is to build a Python-based system that reads student marks and attendance, processes the data, and produces useful summaries. It calculates totals, averages, grades, pass/fail status, subject-wise averages, class statistics, and top performers.

3. Technologies Used

Python; Pandas for reading and organizing CSV data in a DataFrame; NumPy for numerical operations; CSV for storing student records. No machine-learning model is required for this assignment.

4. Dataset Description

The supplied sample dataset contains 20 fictional student records. Each record includes Student ID, Name, Department, marks in Python, Mathematics, AI, and Data Analysis, and Attendance percentage. The data is illustrative and should be replaced or adapted if the trainer requires real or personally collected data.

5. Implementation

1. Read students.csv using pandas.read_csv().

2. Check that required columns exist.

3. Calculate total marks by summing the four subject columns.

4. Calculate average marks by taking the mean of the subject columns.

5. Assign grades using a reusable function.

6. Mark a student as Pass only when every subject mark is at least 40; otherwise mark Fail.

7. Calculate class average, highest and lowest student averages, pass/fail counts, average attendance, and subject-wise averages.

8. Sort students by average and display the top three.

9. Save the processed data to student_analysis_results.csv.

6. Key Features

• Student-wise total and average marks

• Grade assignment

• Pass/fail classification

• Highest, lowest, and overall class average

• Pass/fail counts and average attendance

• Subject-wise performance averages

• Top-three student listing

• Export of calculated results to a CSV file

7. Output / Screenshots

The program prints a dataset preview, student-wise results, class summary, subject-wise averages, and top performers. Run `python main.py` and take screenshots of the terminal output for the final submission. A sample terminal output is included in sample_output.txt.

8. Final Outcome

The project turns a basic CSV dataset into a readable performance report. It demonstrates how Python, Pandas, and NumPy can help organize student data and calculate useful academic summaries.

9. Challenges & Learning

Suggested points to personalize after testing: learning to read CSV files with Pandas; selecting subject columns for calculations; creating reusable functions for grades and pass/fail logic; checking column names; and exporting the final results. Write only challenges you actually faced and how you solved them.

10. GitHub Repository

Repository URL: _____

After uploading the project, paste the actual public repository URL here and submit it through the Google Form provided by EWB Courses.

Grading rules used in this sample

A+: 90 and above; A: 80–89.99; B: 70–79.99; C: 60–69.99; D: 50–59.99; E: 40–49.99; F: below 40. A student passes only if all four subject marks are at least 40. These are project-defined sample rules; follow any different rule specified by your trainer.